

# Power System Analysis – Full Interview Guide

## 1. Two-minute Project Explanation

In this project, I analyzed hourly German electricity system data from 2020 to 2025 to understand demand behavior, generation patterns, and residual load dynamics.

I started by building a structured data processing pipeline in Python. First, I loaded separate generation and consumption datasets, converted timestamps, cleaned numeric columns, standardized column names, and merged both datasets into one consistent hourly table.

After that, I created derived indicators such as renewable generation, fossil generation, total generation, renewable share, and time-based features like year, month, hour, and weekend flags.

In the demand analysis, I examined long-term demand trends, seasonal variation, average hourly demand profiles, and weekday-versus-weekend differences.

In the generation analysis, I looked at the overall generation mix, renewable source breakdown, seasonal wind and solar behavior, and the evolution of renewable share over time.

The most important part of the project was the residual load analysis, because it shows the remaining demand after renewable generation is considered.

There I analyzed the residual load time series, negative residual load hours, duration curves, duck curve behavior, and ramping distribution.

Overall, the project helped me combine Python-based data analysis with energy-system interpretation, especially around renewable integration, flexibility needs, and system balancing.

## 2. One-minute Summary

This project analyzes hourly German power system data from 2020 to 2025 using Python.

I cleaned and merged generation and consumption datasets, created key indicators such as renewable generation and residual load, and then analyzed demand trends, generation mix, and flexibility-related behavior.

The most important insight was how renewable generation reshapes residual load, including negative residual load events, duck curve effects, and ramping needs.

## 3. Key Insights

Electricity demand shows clear seasonal and intraday variation, with lower demand at night and higher demand during daytime and evening.

Renewable generation plays an increasingly important role in the generation mix.

Solar and wind have different seasonal patterns, showing complementarity.

Residual load is strongly affected by renewable generation, especially solar.

Negative residual load hours and strong ramping highlight flexibility needs.

## **4. Why This Project?**

I chose this project because it combines data analysis with energy-system understanding and allows working with real hourly electricity data.

## **5. Technical Challenges**

One challenge was cleaning and standardizing raw time-series data, including timestamps and numeric formats.

Another challenge was structuring the workflow from raw data to meaningful indicators in a clear and reusable way.

## **6. Importance of Residual Load**

Residual load shows how much demand remains after renewable generation and is key for understanding flexibility and balancing requirements.

## **7. Future Improvements**

I would extend the project with forecasting, electricity price integration, and storage modeling.

## **8. Closing Statement**

This project reflects my approach of transforming raw energy data into meaningful insights relevant for real-world energy decisions.